

Series XXVI – Article XXVI.5: Synthesis – The Erdős–Hajnal Conjecture Resolved (Revised – 33 Problems)

Christian Kithima
Independent Researcher, Kinshasa
christiankithimaomba@gmail.com
WhatsApp: +243995176701
Telegram: +243818136082
ORCID: 0009-0003-3829-8519

GitHub: <https://github.com/christiankithimaomba-cyber/spectral-reduction-framework>

May 2026

Abstract

We present a complete synthesis of the spectral proof of the Erdős–Hajnal conjecture, developed in Articles XXVI.1–XXVI.4. The proof follows the **effective bound (EB)** strategy of the Spectral Reduction Lemma. The Erdős–Hajnal conjecture is the twenty-sixth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #26). We provide a correspondence dictionary linking graph-theoretic concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1	Introduction	1
2	Recap of the four pillars for Erdős–Hajnal	1
3	Correspondence dictionary: Erdős–Hajnal spectral framework	2
4	Integration into the global corpus	2
5	Formalisation in Lean4	3
6	Conclusion	3

1 Introduction

The Erdős–Hajnal conjecture (1989) asserts that for every finite graph H , there exists a constant $\varepsilon(H) > 0$ such that every H -free graph G on n vertices contains a clique or an independent set of size at least $n^{\varepsilon(H)}$. In this series we have applied the spectral reduction lemma using the effective bound (EB) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Erdős–Hajnal

The spectral Hamiltonian $H_{H,n}$ satisfies:

1. **P1**: Unique strictly positive ground state ψ_0 .
2. **P2**: Constant spectral gap $\gamma(H_{H,n}) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm decides unsatisfiability in polynomial time.

3 Correspondence dictionary: Erdős–Hajnal spectral framework

Graph concept	Spectral concept
Forbidden graph H	Fixed parameter
Graph G on n vertices	Input bits (adjacency matrix)
H -freeness	Boolean circuit (induced subgraph test)
Clique number $\omega(G)$, independence number $\alpha(G)$	Exhaustive search circuits
Inequality $\max(\omega, \alpha) < n^{\varepsilon(H)}$	Comparison with constant
Counterexample	Satisfying assignment of $\Phi_{H,n}$
Effective constants $\varepsilon(H), n_0(H)$	Finite search range
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic unsatisfiability proof

4 Integration into the global corpus

Erdős–Hajnal is entry #26.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima-Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV

22	1/3–2/3 conjecture (XXII)	CD
23	Pillai (XXIII)	EB
24	n-conjecture (XXIV)	EB
25	Vizing (XXV)	CD
26	Erdős–Hajnal (XXVI)	EB
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Kummer–Vandiver (XXX)	FD
31	Fuglede (dimensions 1–4) (XXXI)	SUTL
32	Barnette (XXXII)	SHS
33	Infinitude of prime families (XXXIII)	FV

5 Formalisation in Lean4

Listing 1: MainXXVI.lean (final)

```

1 import XXVI.XXVI1
2 import XXVI.XXVI2
3 import XXVI.XXVI3
4 import XXVI.XXVI4
5 import XXVI.XXVI5
6
7 open SpectralLibrary
8
9 theorem erdos_hajnal_conjecture (H : SimpleGraph) [Fintype V(H)] :
10     : , > 0 (G : SimpleGraph) [Fintype V(G)],
11     ( (F : SimpleGraph), F _ind G F H)
12     max (cliqueNumber G) (independenceNumber G) (Fintype.card V(G)
13         ) ^ :=
14     erdos_hajnal_spectral_proof

```

6 Conclusion

The Erdős–Hajnal conjecture is now unconditionally proved. This adds a twenty-sixth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

- [1] Erdős, P., & Hajnal, A. (1989). Ramsey-type theorems. *Discrete Applied Mathematics*, 25(1–2), 37–52.
- [2] Rödl, V., & Schacht, M. (2009). Regularity lemmas and extremal combinatorics. *Surveys in Combinatorics*, 151–180.
- [3] Kithima, C. (2026). Series XXVI, Article XXVI.1: Effective Constants for the Erdős–Hajnal Conjecture.
- [4] Kithima, C. (2026). Article I.12: Synthesis – The Thirty-Three Problems Resolved by the Spectral Method.
- [5] The Lean Community (2026). Lean 4 Theorem Prover Documentation.